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Why Use Beamer?

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- This template applies NTU Blue to the title bars, navigation, and footer.
- The supplied logo is loaded from `pic/NTU_Logo.png`.
- Compile with $\text{\XeLaTeX}$ or $\text{\LuaLaTeX}$ when the deck contains Chinese text.

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English and Chinese

English is the default presentation language. The template also loads `ctex` when compiled with `Xe $\LaTeX$` or `Lua $\LaTeX$` , so Chinese can be used wherever it is needed.

Chinese example

南洋理工大学欢迎您。这个模板支持中英文混合排版。

Source

`\kaishu` may be applied locally to Chinese text; it is no longer applied to the entire English presentation.

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Theme Features

- A compact navigation bar keeps section markers on one line.
- NTU Blue is the primary color throughout the presentation.
- The footer displays the author, institution, title, and frame count.
- Numbered captions and text-style bibliography entries are enabled.
- The theme remains easy to extend for courses, conferences, and research groups.
- The sample bibliography retains a reference to the original theme [unk15].

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Document Preparation

Word processor

Direct visual editing
Manual formatting is common
Long documents require careful maintenance
Equation tools vary by application
Usually stored in a binary format

Often requires a commercial licence

L^AT_EX

Content-driven typesetting
Styles remain consistent
Long and short documents use the same workflow
Mathematical typesetting is a core strength
Source files are readable plain text
Free and open-source tooling is available

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Equation Layout

Unnumbered equation

$$J(\theta) = \mathbb{E}_{\pi_{\theta}}[G_t] = \sum_{s \in \mathcal{S}} d^{\pi}(s) \sum_{a \in \mathcal{A}} \pi_{\theta}(a|s) Q^{\pi}(s, a)$$

Aligned equations

Use `\mathrm{}` or `\text{}` for words inside mathematics.

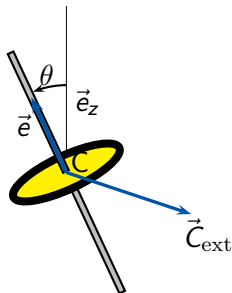
$$\begin{aligned} Q_{\text{target}} &= r + \gamma Q^{\pi}(s', \pi_{\theta}(s') + \epsilon), \\ \epsilon &\sim \text{clip}(\mathcal{N}(0, \sigma), -c, c). \end{aligned} \tag{1}$$

Numbered Multiline Equation

Multiline environment

$$\begin{aligned} A = \lim_{n \rightarrow \infty} \Delta x & \left(a^2 + \left(a^2 + 2a\Delta x + (\Delta x)^2 \right) \right. \\ & + \left(a^2 + 2 \cdot 2a\Delta x + 2^2 (\Delta x)^2 \right) + \dots \\ & \left. + \left(a^2 + 2 \cdot (n-1)a\Delta x + (n-1)^2 (\Delta x)^2 \right) \right) \\ & = \frac{1}{3} (b^3 - a^3). \quad (2) \end{aligned}$$

Graphics and Columns



1	2	3	A	697 Hz
4	5	6	B	770 Hz
7	8	9	C	852 Hz
*	0	#	D	941 Hz
1209 Hz	1366 Hz	1477 Hz	1633 Hz	

Figure 1: A second figure in an adjacent column

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Common L^AT_EX Commands

Commands

<code>\chapter</code>	<code>\section</code>	<code>\subsection</code>	<code>\paragraph</code>
Chapter	Section	Subsection	Titled paragraph
<code>\centering</code>	<code>\emph</code>	<code>\verb</code>	<code>\url</code>
Centre	Emphasise	Verbatim text	Hyperlink
<code>\footnote</code>	<code>\item</code>	<code>\caption</code>	<code>\includegraphics</code>
Footnote	List item	Caption	Insert image
<code>\label</code>	<code>\cite</code>	<code>\ref</code>	
Label	Cite a source	Cross-reference	

Environments

<code>table</code>	<code>figure</code>	<code>equation</code>
Table	Figure	Equation
<code>itemize</code>	<code>enumerate</code>	<code>description</code>
Bulleted list	Numbered list	Description list

List Environments

```
1 \begin{itemize}
2   \item First item
3   \item Second item
4   \begin{itemize}
5     \item Nested item
6   \end{itemize}
7 \end{itemize}
```

- First item
- Second item
 - Nested item

List Environments

```
1 \begin{itemize}
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6   \end{itemize}
7 \end{itemize}
```

- First item
- Second item
 - Nested item

```
1 \begin{enumerate}
2   \item Beginner
3   \item Intermediate
4   \item Advanced
5 \end{enumerate}
```

- ① Beginner
- ② Intermediate
- ③ Advanced

Mathematical Expressions

```
1 $V = \frac{4}{3}\pi r^3$  
2  
3 \[  
4   V = \frac{4}{3}\pi r^3  
5 \]  
6  
7 \begin{equation}  
8   \label{eq:vsphere}  
9   V = \frac{4}{3}\pi r^3  
10 \end{equation}
```

Inline: $V = \frac{4}{3}\pi r^3$

$$V = \frac{4}{3}\pi r^3$$

$$V = \frac{4}{3}\pi r^3 \quad (3)$$

- See the [L^AT_EX mathematics guide](#) for more examples.

Tables and Cross-References

```
1 \begin{table}
2   \caption{Number and value}
3   \label{tab:number}
4   \begin{tabular}{cl}
5     \toprule
6     Number & Value \\
7     \midrule
8     1 & 4.0 \\
9     2 & 3.7 \\
10    \bottomrule
11    \end{tabular}
12 \end{table}
13 See Equation~\ref{eq:vsphere}
14 and Table~\ref{tab:number}.
```

Table 1: Number and value

Number	Value
1	4.0
2	3.7

See Equation (3) and
Table 1.

Choosing Graphic Formats

- Prefer vector graphics such as PDF, EPS, and PS for diagrams.
 - Useful tools include PGF/TikZ, PSTricks, Inkscape, and Visio.
 - Charts from MATLAB or Excel can be exported as PDF.
- Use raster graphics such as PNG, JPEG, and TIFF for photographs.
 - Export at sufficient resolution to avoid visible blur.
 - Avoid raster formats for line art when a vector source is available.



Figure 2: NTU logo example

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Project Timeline

- January: complete the literature review.
- February: evaluate candidate presentation structures.
- March and April: develop and test the NTU Beamer theme.
- May: prepare the final report and presentation.

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References I

[unk15] unknown.

Original beamer theme reference, 2015.

Reference kept for the sample bibliography frame.

Thank you!